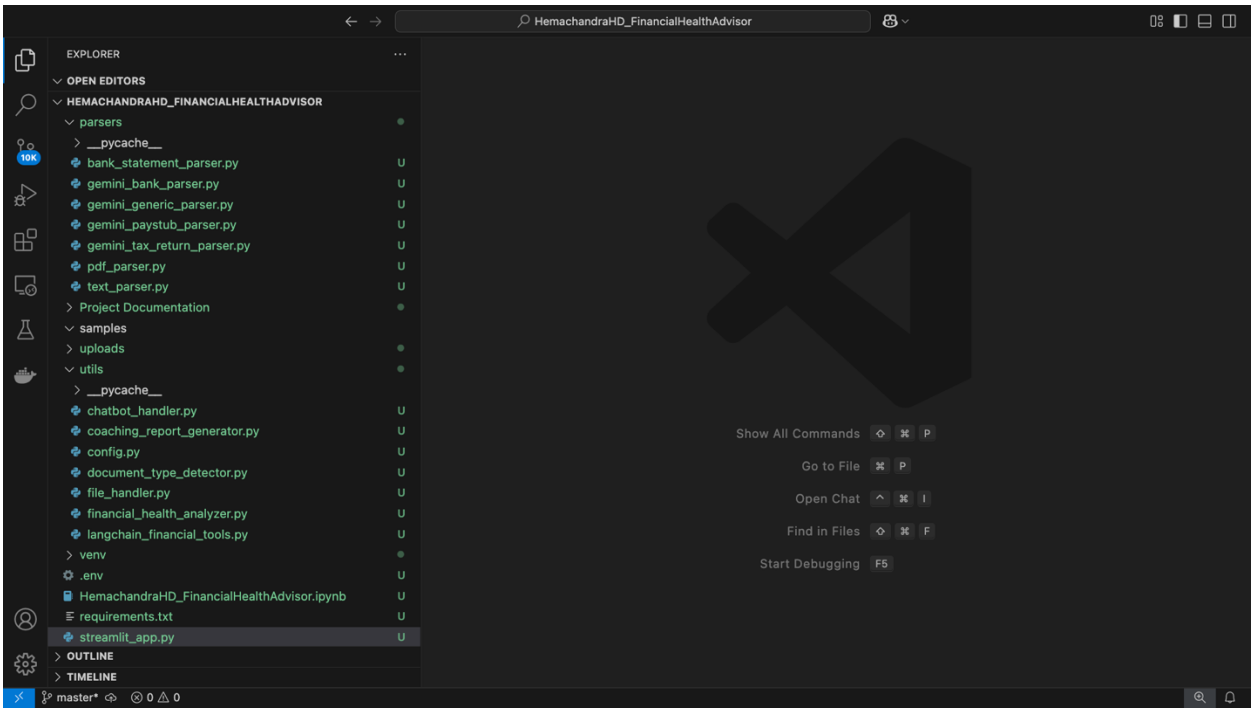
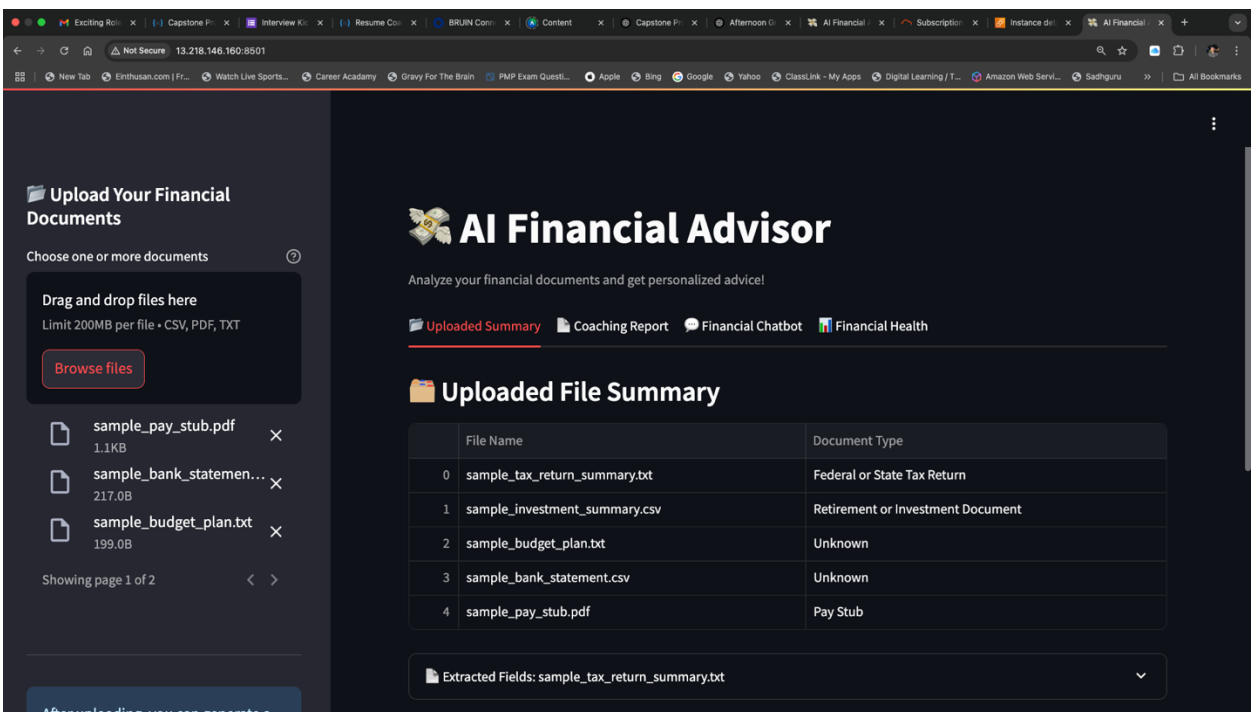
Application & Deployment Screen Shots:
AI-Powered Financial Health Advisor Capstone Project

HEMACHANDRA HD

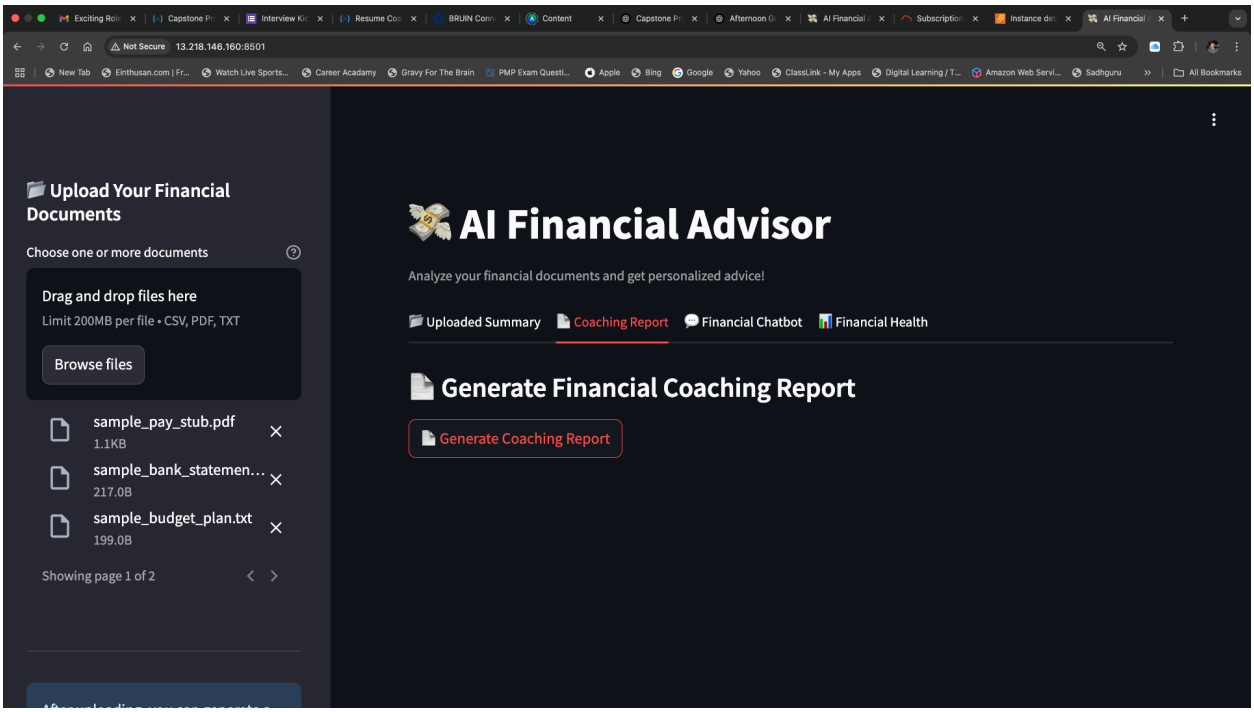
Project folder Structure:

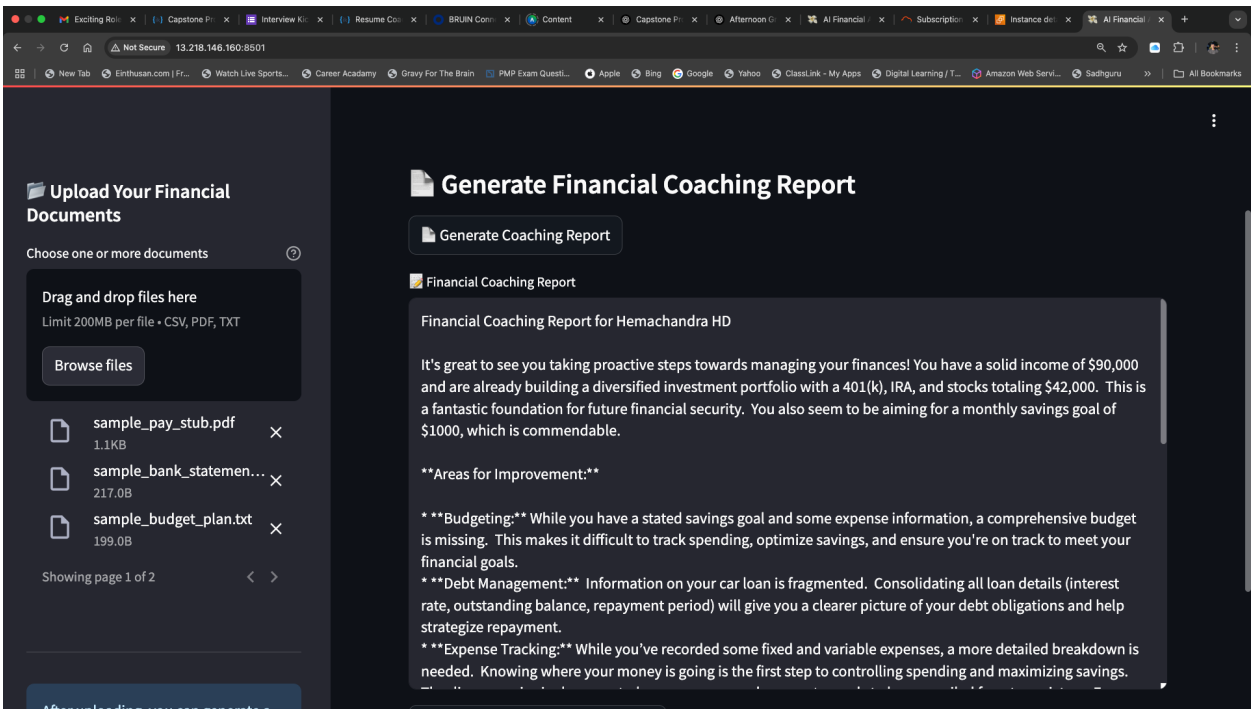


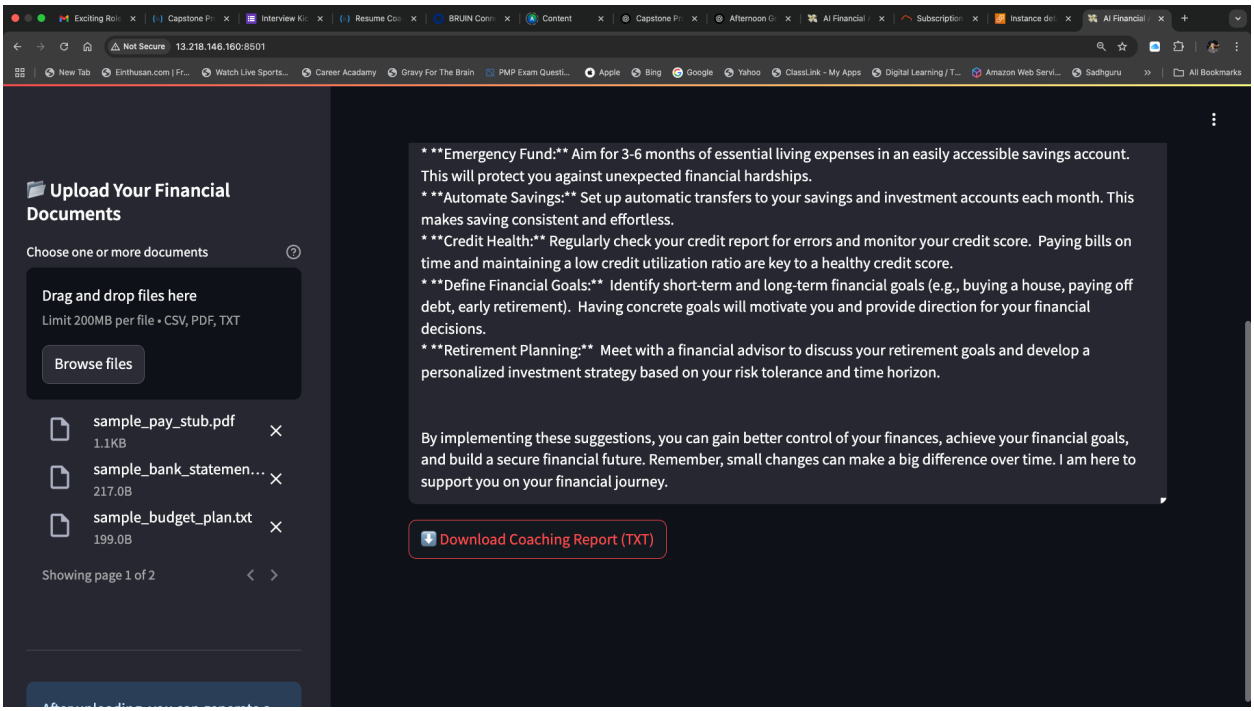
Streamlit-based web UI allowing users to upload paystubs, bank statements, and tax returns



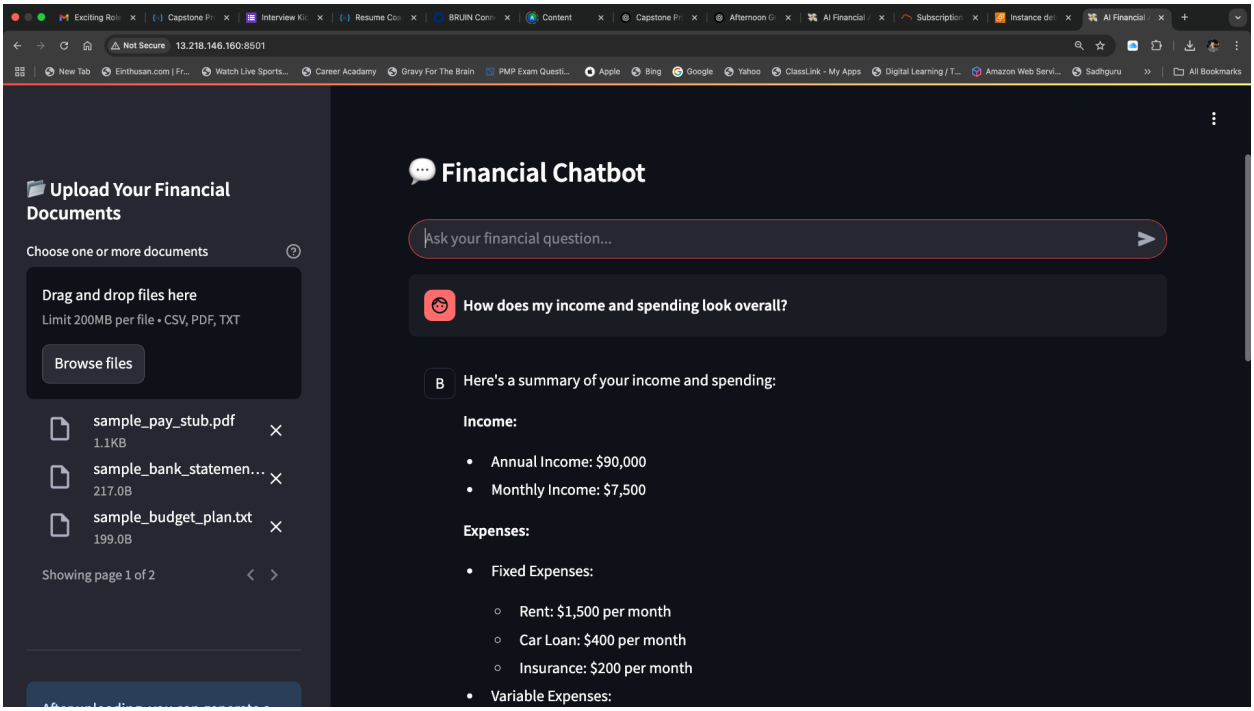
Streamlit-based web UI – Coaching report

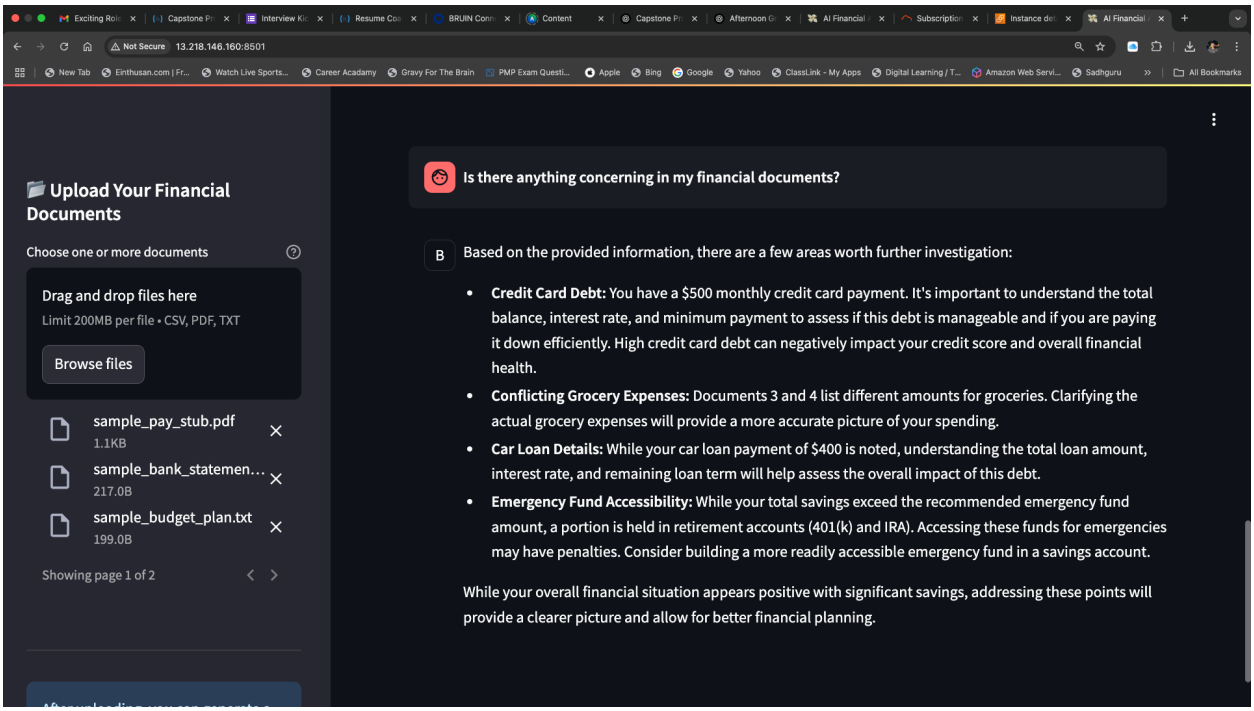




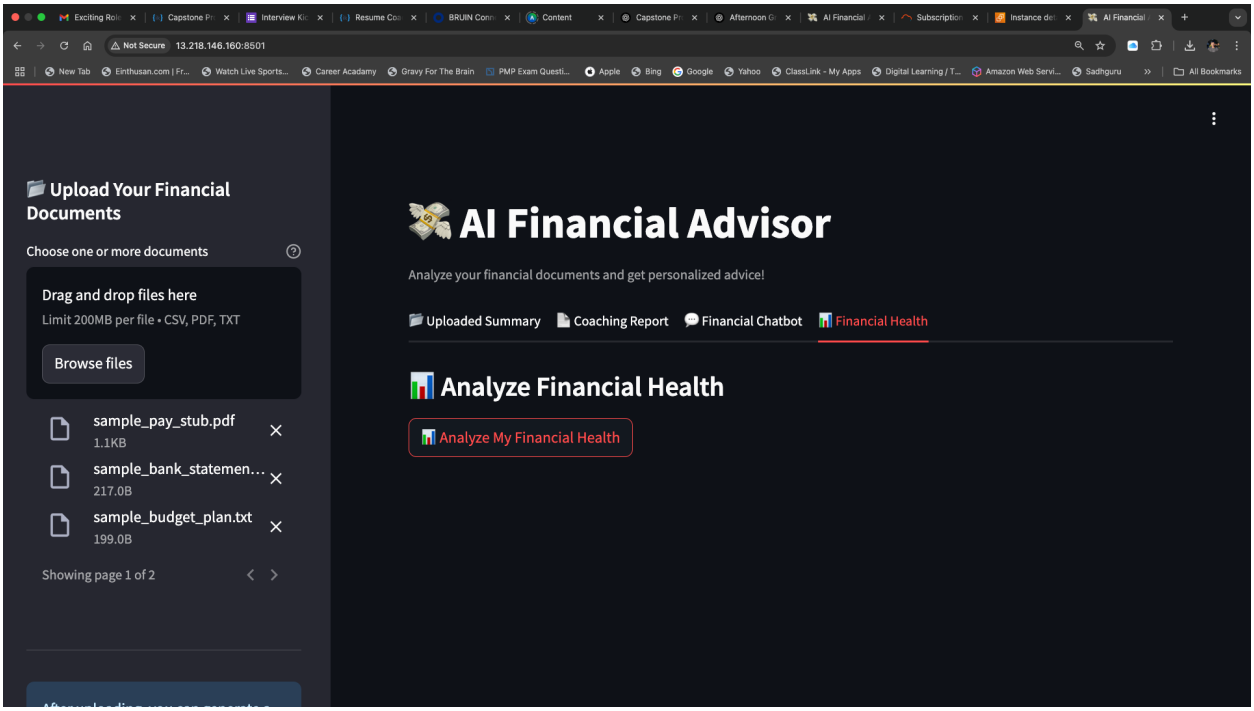


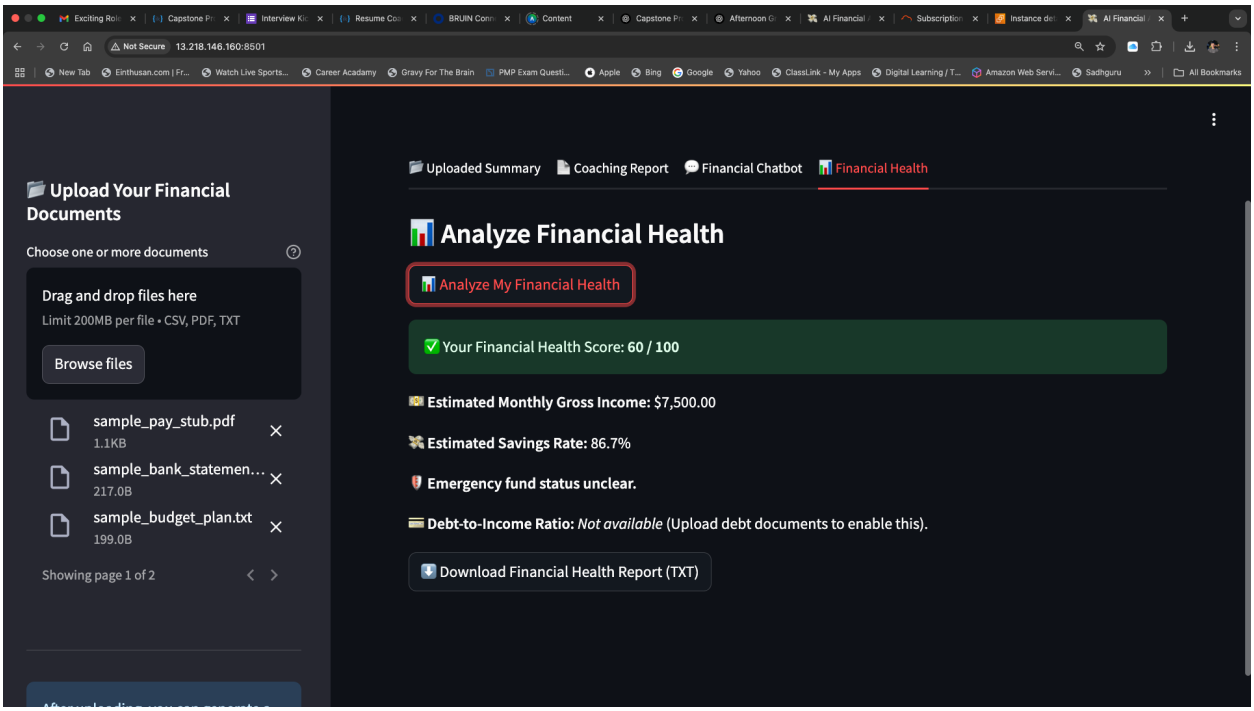
Streamlit-based web UI – Financial Chatbot



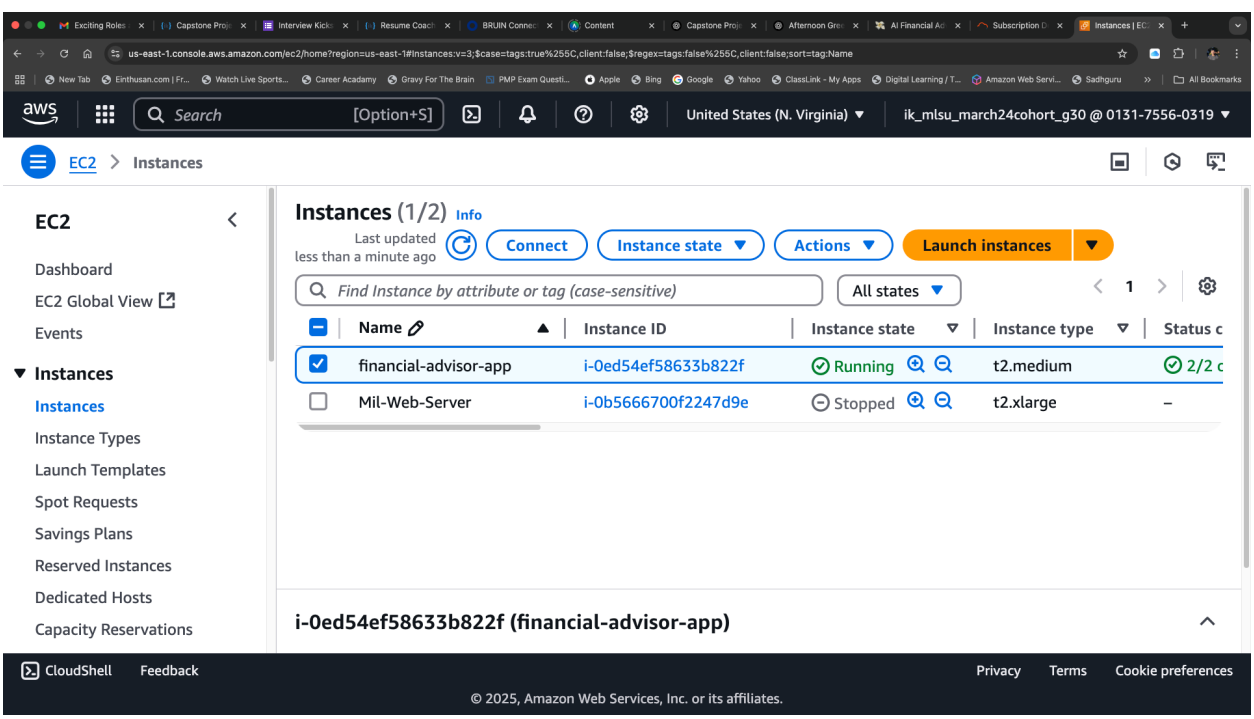


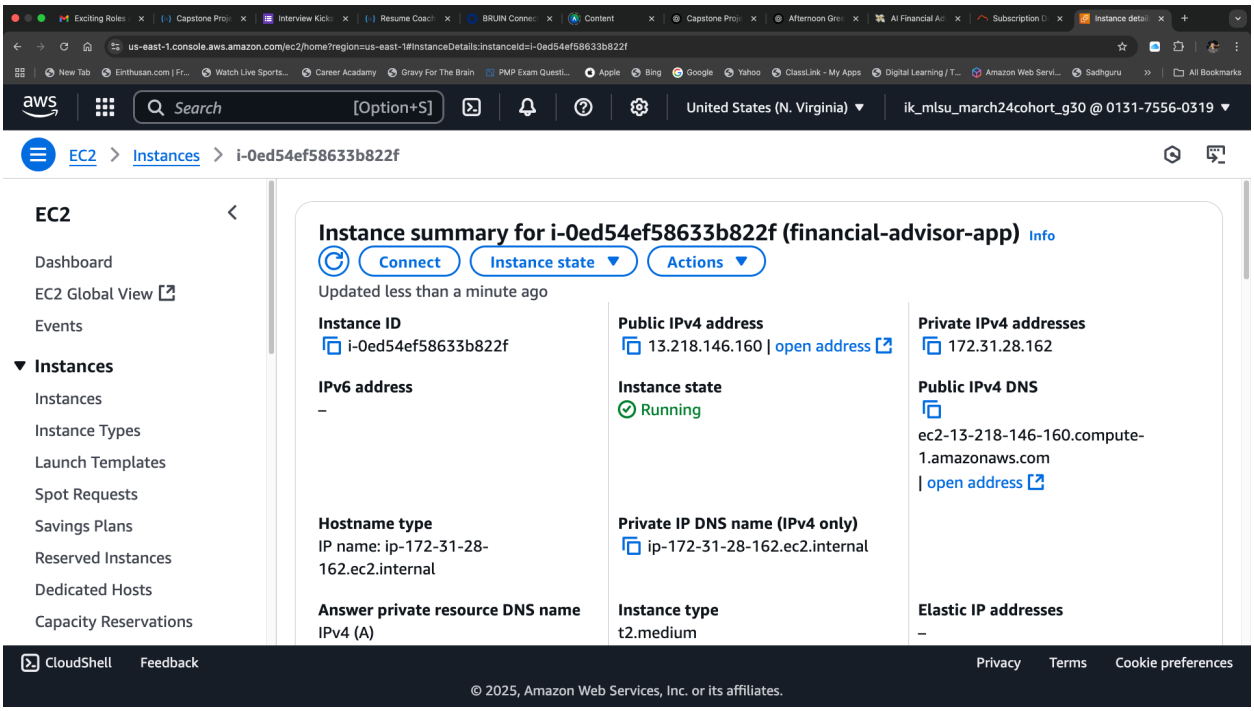
Streamlit-based web UI – Financial Health Check

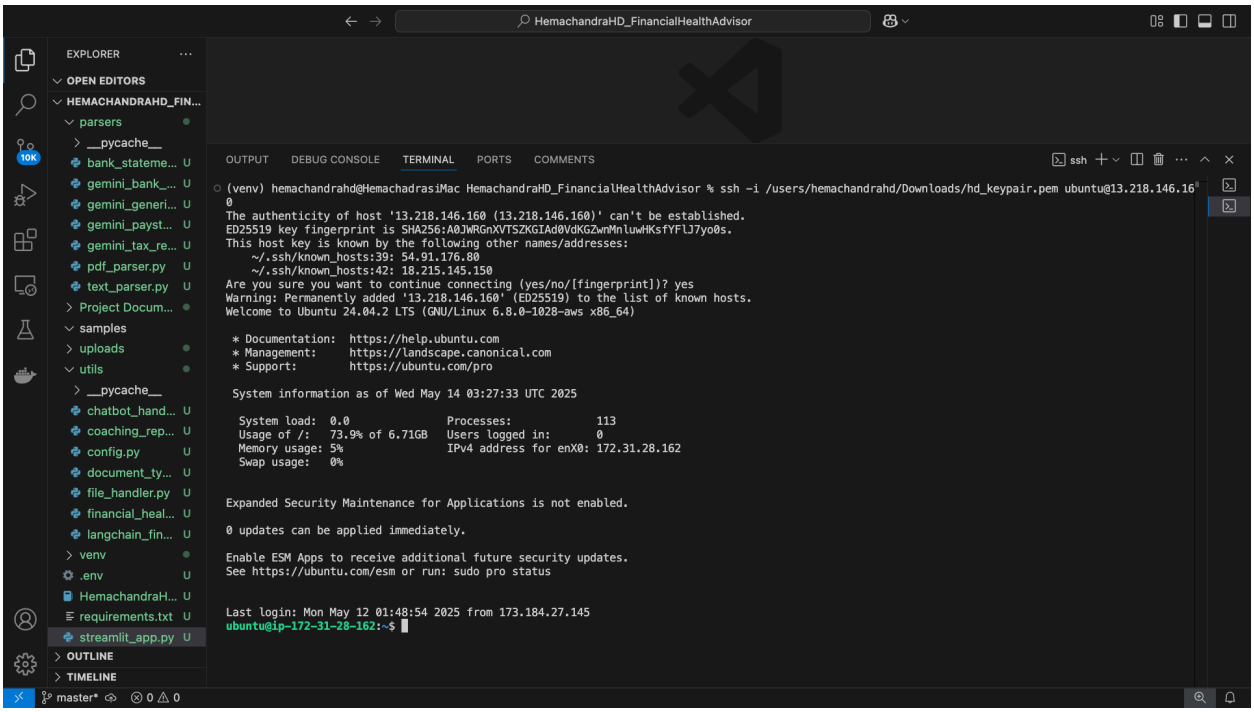




AWS Cloud Deployment:




us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#InstanceDetails:instanceId=i-0ed54ef58633b822f
aws
Search
[Option+S]
United States (N. Virginia)
ik_mlsu_march24cohort_g30 @ 0131-7556-0319
EC2 > Instances > i-0ed54ef58633b822f
EC2
Dashboard
EC2 Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Instance summary for i-0ed54ef58633b822f (financial-advisor-app) Info
Connect
Instance state
Actions
Updated less than a minute ago
Instance ID
i-0ed54ef58633b822f
Public IPv4 address
13.218.146.160 | open address
Private IPv4 addresses
172.31.28.162
IPv6 address
–
Instance state
Running
Public IPv4 DNS
ec2-13-218-146-160.compute-1.amazonaws.com
| open address
Hostname type
IP name: ip-172-31-28-162.ec2.internal
Private IP DNS name (IPv4 only)
ip-172-31-28-162.ec2.internal
Answer private resource DNS name
IPv4 (A)
Instance type
t2.medium
Elastic IP addresses
–
CloudShell
Feedback
Privacy
Terms
Cookie preferences
© 2025, Amazon Web Services, Inc. or its affiliates.


HemachandraHD_FinancialHealthAdvisor
EXPLORER
OPEN EDITORS
HEMACHANDRAHD_FIN...
parsers
__pycache__
bank_stateme...
gemini_bank_...
gemini_generi...
gemini_payst...
gemini_tax_re...
pdf_parser.py
text_parser.py
Project Docum...
samples
uploads
utils
__pycache__
chatbot_hand...
coaching_rep...
config.py
document_ty...
file_handler.py
financial_heal...
langchain_fin...
venv
.env
HemachandraH...
requirements.txt
streamlit_app.py
OUTLINE
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TERMINAL
PORTS
COMMENTS
ssh
(venv) hemachandrahd@HemachadrasiMac HemachandraHD_FinancialHealthAdvisor % ssh -i /users/hemachandrahd/Downloads/hd_keypair.pem ubuntu@13.218.146.160
The authenticity of host '13.218.146.160 (13.218.146.160)' can't be established.
ED25519 key fingerprint is SHA256:A0JWRGnXVTSZKGIAd0VdKGZwnMnluwHKsfYFlJ7yo0s.
This host key is known by the following other names/addresses:
~/.ssh/known_hosts:39: 54.91.176.80
~/.ssh/known_hosts:42: 18.215.145.150
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '13.218.146.160' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.2 LTS (GNU/Linux 6.8.0-1028-aws x86_64)
* Documentation: https://help.ubuntu.com
* Management: https://landscape.canonical.com
* Support: https://ubuntu.com/pro
System information as of Wed May 14 03:27:33 UTC 2025
System load: 0.0
Usage of /: 73.9% of 6.71GB
Memory usage: 5%
Swap usage: 0%
Processes: 113
Users logged in: 0
IPv4 address for enX0: 172.31.28.162
Expanded Security Maintenance for Applications is not enabled.
0 updates can be applied immediately.
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
Last login: Mon May 12 01:48:54 2025 from 173.184.27.145
ubuntu@ip-172-31-28-162:~$
master*

HemachandraHD_FinancialHealthAdvisor

EXPLORER

OPEN EDITORS

HEMACHANDRAHD_FIN...

- parsers
  - __pycache__
  - bank_stateme... U
  - gemini_bank_... U
  - gemini_generi... U
  - gemini_payst... U
  - gemini_tax_re... U
  - pdf_parser.py U
  - text_parser.py U
  - Project Docum...
- samples
- uploads
- utils
  - __pycache__
  - chatbot_hand... U
  - coaching_rep... U
  - config.py U
  - document_ty... U
  - file_handler.py U
  - financial_heal... U
  - langchain_fin... U
- venv
- .env U
- HemachandraH... U
- requirements.txt U
- streamlit_app.py U

OUTLINE

TIMELINE

OUTPUT DEBUG CONSOLE TERMINAL PORTS COMMENTS

```
(venv) hemachandrahd@HemachadrasiMac HemachandraHD_FinancialHealthAdvisor % ssh -i ~/Downloads/hd_keypair.pem ubuntu@13.218.146.160
Welcome to Ubuntu 24.04.2 LTS (GNU/Linux 6.8.0-1028-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Wed May 14 03:51:56 UTC 2025

  System load:  0.08              Processes:             117
  Usage of /:   79.4% of 6.71GB   Users logged in:       0
  Memory usage: 6%                IPv4 address for enX0: 172.31.28.162
  Swap usage:   0%


Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


Last login: Wed May 14 03:27:34 2025 from 173.184.27.145
ubuntu@ip-172-31-28-162:~$ ls
HemachandraHD_FinancialHealthAdvisor  HemachandraHD_FinancialHealthAdvisor.zip  IK_Capstone_Project  IK_Capstone_Project.zip  __MACOSX  venv
ubuntu@ip-172-31-28-162:~$ unzip HemachandraHD_FinancialHealthAdvisor.zip
Archive:  HemachandraHD_FinancialHealthAdvisor.zip
replace HemachandraHD_FinancialHealthAdvisor/Project Documentation/~$ Powered Financial Advisor - Project Architecture Overview.docx? [y]es, [n]o, [A
]ll, [N]one, [r]ename: y
  inflating: HemachandraHD_FinancialHealthAdvisor/Project Documentation/~$ Powered Financial Advisor - Project Architecture Overview.docx
replace HemachandraHD_FinancialHealthAdvisor/Project Documentation/AI Powered Financial Advisor - User Manual.docx? [y]es, [n]o, [A]ll, [N]one, [r]en
ame: A
  inflating: HemachandraHD_FinancialHealthAdvisor/Project Documentation/AI Powered Financial Advisor - User Manual.docx
  inflating: HemachandraHD_FinancialHealthAdvisor/Project Documentation/AI Powered Financial Advisor - Project Design.docx
  inflating: HemachandraHD_FinancialHealthAdvisor/Project Documentation/AI Powered Financial Advisor - Project Architecture Overview.docx
  inflating: HemachandraHD_FinancialHealthAdvisor/Project Documentation/~$ Powered Financial Advisor - Project Design.docx
  inflating: HemachandraHD_FinancialHealthAdvisor/Project Documentation/~$ Powered Financial Advisor - User Manual.docx
  inflating: HemachandraHD_FinancialHealthAdvisor/Project Documentation/~$ Powered Financial Advisor - Application and Deployment snapshots.docx
  inflating: HemachandraHD_FinancialHealthAdvisor/Project Documentation/AI Powered Financial Advisor - Application and Deployment snapshots.docx
```

master* ⊗ 0 ⚠ 0

HemachandraHD_FinancialHealthAdvisor

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  - langchain_fin... U
- venv
- .env U
- HemachandraH... U
- requirements.txt U
- streamlit_app.py U

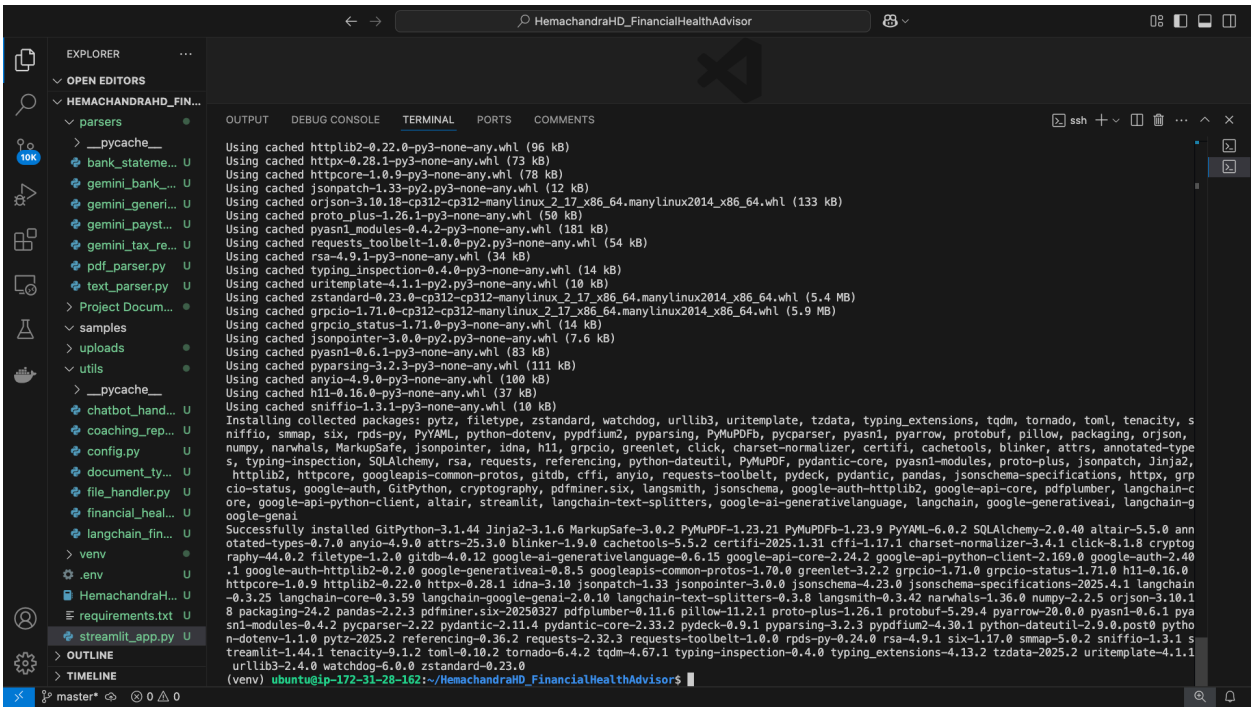
OUTLINE

TIMELINE

OUTPUT DEBUG CONSOLE TERMINAL PORTS COMMENTS

```
ubuntu@ip-172-31-28-162:~/HemachandraHD_FinancialHealthAdvisor$ source venv/bin/activate
(venv) ubuntu@ip-172-31-28-162:~/HemachandraHD_FinancialHealthAdvisor$ pip install -r requirements.txt
Collecting altair==5.5.0 (from -r requirements.txt (line 1))
  Using cached altair-5.5.0-py3-none-any.whl.metadata (11 kB)
Collecting attrs==25.3.0 (from -r requirements.txt (line 2))
  Using cached attrs-25.3.0-py3-none-any.whl.metadata (10 kB)
Collecting blinker==1.9.0 (from -r requirements.txt (line 3))
  Using cached blinker-1.9.0-py3-none-any.whl.metadata (1.6 kB)
Collecting cachetools==5.5.2 (from -r requirements.txt (line 4))
  Using cached cachetools-5.5.2-py3-none-any.whl.metadata (5.4 kB)
Collecting certifi==2025.1.31 (from -r requirements.txt (line 5))
  Using cached certifi-2025.1.31-py3-none-any.whl.metadata (2.5 kB)
Collecting cffi==1.17.1 (from -r requirements.txt (line 6))
  Using cached cffi-1.17.1-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (1.5 kB)
Collecting charset-normalizer==3.4.1 (from -r requirements.txt (line 7))
  Using cached charset_normalizer-3.4.1-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (35 kB)
Collecting click==8.1.8 (from -r requirements.txt (line 8))
  Using cached click-8.1.8-py3-none-any.whl.metadata (2.3 kB)
Collecting cryptography==44.0.2 (from -r requirements.txt (line 9))
  Using cached cryptography-44.0.2-cp39-abi3-manylinux_2_34_x86_64.whl.metadata (5.7 kB)
Collecting gitdb==4.0.12 (from -r requirements.txt (line 10))
  Using cached gitdb-4.0.12-py3-none-any.whl.metadata (1.2 kB)
Collecting GitPython==3.1.44 (from -r requirements.txt (line 11))
  Using cached GitPython-3.1.44-py3-none-any.whl.metadata (13 kB)
Collecting idna==3.10 (from -r requirements.txt (line 12))
  Using cached idna-3.10-py3-none-any.whl.metadata (10 kB)
Collecting Jinja2==3.1.6 (from -r requirements.txt (line 13))
  Using cached jinja2-3.1.6-py3-none-any.whl.metadata (2.9 kB)
Collecting jsonschema==4.23.0 (from -r requirements.txt (line 14))
  Using cached jsonschema-4.23.0-py3-none-any.whl.metadata (7.9 kB)
Collecting jsonschema-specifications==2025.4.1 (from -r requirements.txt (line 15))
  Using cached jsonschema_specifications-2025.4.1-py3-none-any.whl.metadata (2.9 kB)
Collecting MarkupSafe==3.0.2 (from -r requirements.txt (line 16))
  Using cached MarkupSafe-3.0.2-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (4.0 kB)
Collecting narwhals==1.36.0 (from -r requirements.txt (line 17))
  Using cached narwhals-1.36.0-py3-none-any.whl.metadata (9.2 kB)
Collecting numpy==2.2.5 (from -r requirements.txt (line 18))
  Using cached numpy-2.2.5-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (62 kB)
Collecting packaging==24.2 (from -r requirements.txt (line 19))
  Using cached packaging-24.2-py3-none-any.whl.metadata (3.2 kB)
```

master* ⊗ 0 ⚠ 0


HemachandraHD_FinancialHealthAdvisor
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OPEN EDITORS
HEMACHANDRAHD_FIN...
parsers
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gemini_bank_... U
gemini_generi... U
gemini_payst... U
gemini_tax_re... U
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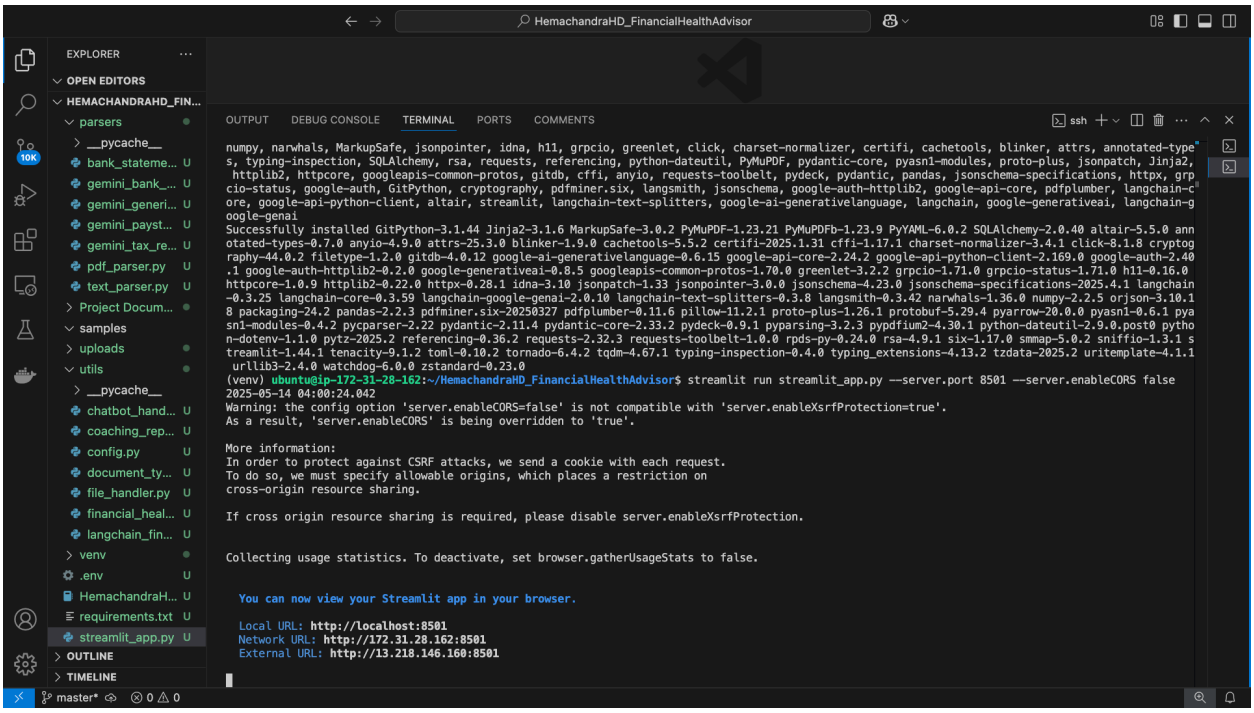
file_handler.py U
financial_heal... U
langchain_fin... U
venv
.env U
HemachandraH... U
requirements.txt U
streamlit_app.py U
OUTLINE
TIMELINE
OUTPUT DEBUG CONSOLE TERMINAL PORTS COMMENTS
ssh
Using cached httplib2-0.22.0-py3-none-any.whl (96 kB)
Using cached httpx-0.28.1-py3-none-any.whl (73 kB)
Using cached httpcore-1.0.9-py3-none-any.whl (78 kB)
Using cached jsonpatch-1.33-py2.py3-none-any.whl (12 kB)
Using cached orjson-3.10.18-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (133 kB)
Using cached proto_plus-1.26.1-py3-none-any.whl (50 kB)
Using cached pyasn1_modules-0.4.2-py3-none-any.whl (181 kB)
Using cached requests_toolbelt-1.0.0-py2.py3-none-any.whl (54 kB)
Using cached rsa-4.9.1-py3-none-any.whl (34 kB)
Using cached typing_inspection-0.4.0-py3-none-any.whl (14 kB)
Using cached uritemplate-4.1.1-py2.py3-none-any.whl (10 kB)
Using cached zstandard-0.23.0-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (5.4 MB)
Using cached grpcio-1.71.0-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (5.9 MB)
Using cached grpcio_status-1.71.0-py3-none-any.whl (14 kB)
Using cached jsonpointer-3.0.0-py2.py3-none-any.whl (7.6 kB)
Using cached pyasn1-0.6.1-py3-none-any.whl (83 kB)
Using cached pyparsing-3.2.3-py3-none-any.whl (111 kB)
Using cached anyio-4.9.0-py3-none-any.whl (100 kB)
Using cached h11-0.16.0-py3-none-any.whl (37 kB)
Using cached sniffio-1.3.1-py3-none-any.whl (10 kB)
Installing collected packages: pytz, filetype, zstandard, watchdog, urllib3, uritemplate, tzdata, typing_extensions, tqdm, tornado, toml, tenacity, s
niffio, smmap, six, rpds-py, PyYAML, python-dotenv, pypdfium2, pyparsing, PyMuPDFb, pycparser, pyasn1, pyarrow, protobuf, pillow, packaging, orjson,
numpy, narwhals, MarkupSafe, jsonpointer, idna, h11, grpcio, greenlet, click, charset-normalizer, certifi, cachetools, blinker, attrs, annotated-type
s, typing-inspection, SQLAlchemy, rsa, requests, referencing, python-dateutil, PyMuPDF, pydantic-core, pyasn1-modules, proto-plus, jsonpatch, Jinja2,
 httplib2, httpcore, googleapis-common-protos, gitdb, cffi, anyio, requests-toolbelt, pydeck, pydantic, pandas, jsonschema-specifications, httpx, grp
cio-status, google-auth, GitPython, cryptography, pdfminer.six, langsmith, jsonschema, google-auth-httplib2, google-api-core, pdfplumber, langchain-c
ore, google-api-python-client, altair, streamlit, langchain-text-splitters, google-ai-generativelanguage, langchain, google-generativeai, langchain-g
oogle-genai
Successfully installed GitPython-3.1.44 Jinja2-3.1.6 MarkupSafe-3.0.2 PyMuPDF-1.23.21 PyMuPDFb-1.23.9 PyYAML-6.0.2 SQLAlchemy-2.0.40 altair-5.5.0 ann
otated-types-0.7.0 anyio-4.9.0 attrs-25.3.0 blinker-1.9.0 cachetools-5.5.2 certifi-2025.1.31 cffi-1.17.1 charset-normalizer-3.4.1 click-8.1.8 cryptog
raphy-44.0.2 filetype-1.2.0 gitdb-4.0.12 google-ai-generativelanguage-0.6.15 google-api-core-2.24.2 google-api-python-client-2.169.0 google-auth-2.40
.1 google-auth-httplib2-0.2.0 google-generativeai-0.8.5 googleapis-common-protos-1.70.0 greenlet-3.2.2 grpcio-1.71.0 grpcio-status-1.71.0 h11-0.16.0
httpcore-1.0.9 httplib2-0.22.0 httpx-0.28.1 idna-3.10 jsonpatch-1.33 jsonpointer-3.0.0 jsonschema-4.23.0 jsonschema-specifications-2025.4.1 langchain
-0.3.25 langchain-core-0.3.59 langchain-google-genai-2.0.10 langchain-text-splitters-0.3.8 langsmith-0.3.42 narwhals-1.36.0 numpy-2.2.5 orjson-3.10.1
8 packaging-24.2 pandas-2.2.3 pdfminer.six-20250327 pdfplumber-0.11.6 pillow-11.2.1 proto-plus-1.26.1 protobuf-5.29.4 pyarrow-20.0.0 pyasn1-0.6.1 pya
sn1-modules-0.4.2 pycparser-2.22 pydantic-2.11.4 pydantic-core-2.33.2 pydeck-0.9.1 pyparsing-3.2.3 pypdfium2-4.30.1 python-dateutil-2.9.0.post0 pytho
n-dotenv-1.1.0 pytz-2025.2 referencing-0.36.2 requests-2.32.3 requests-toolbelt-1.0.0 rpds-py-0.24.0 rsa-4.9.1 six-1.17.0 smmap-5.0.2 sniffio-1.3.1 s
treamlit-1.44.1 tenacity-9.1.2 toml-0.10.2 tornado-6.4.2 tqdm-4.67.1 typing-inspection-0.4.0 typing_extensions-4.13.2 tzdata-2025.2 uritemplate-4.1.1
 urllib3-2.4.0 watchdog-6.0.0 zstandard-0.23.0
(venv) ubuntu@ip-172-31-28-162:~/HemachandraHD_FinancialHealthAdvisor$
master*
0 0


HemachandraHD_FinancialHealthAdvisor
EXPLORER
OPEN EDITORS
HEMACHANDRAHD_FIN...
parsers
__pycache__
bank_stateme... U
gemini_bank_... U
gemini_generi... U
gemini_payst... U
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HemachandraH... U
requirements.txt U
streamlit_app.py U
OUTLINE
TIMELINE
OUTPUT DEBUG CONSOLE TERMINAL PORTS COMMENTS
ssh
numpy, narwhals, MarkupSafe, jsonpointer, idna, h11, grpcio, greenlet, click, charset-normalizer, certifi, cachetools, blinker, attrs, annotated-type
s, typing-inspection, SQLAlchemy, rsa, requests, referencing, python-dateutil, PyMuPDF, pydantic-core, pyasn1-modules, proto-plus, jsonpatch, Jinja2,
 httplib2, httpcore, googleapis-common-protos, gitdb, cffi, anyio, requests-toolbelt, pydeck, pydantic, pandas, jsonschema-specifications, httpx, grp
cio-status, google-auth, GitPython, cryptography, pdfminer.six, langsmith, jsonschema, google-auth-httplib2, google-api-core, pdfplumber, langchain-c
ore, google-api-python-client, altair, streamlit, langchain-text-splitters, google-ai-generativelanguage, langchain, google-generativeai, langchain-g
oogle-genai
Successfully installed GitPython-3.1.44 Jinja2-3.1.6 MarkupSafe-3.0.2 PyMuPDF-1.23.21 PyMuPDFb-1.23.9 PyYAML-6.0.2 SQLAlchemy-2.0.40 altair-5.5.0 ann
otated-types-0.7.0 anyio-4.9.0 attrs-25.3.0 blinker-1.9.0 cachetools-5.5.2 certifi-2025.1.31 cffi-1.17.1 charset-normalizer-3.4.1 click-8.1.8 cryptog
raphy-44.0.2 filetype-1.2.0 gitdb-4.0.12 google-ai-generativelanguage-0.6.15 google-api-core-2.24.2 google-api-python-client-2.169.0 google-auth-2.40
.1 google-auth-httplib2-0.2.0 google-generativeai-0.8.5 googleapis-common-protos-1.70.0 greenlet-3.2.2 grpcio-1.71.0 grpcio-status-1.71.0 h11-0.16.0
httpcore-1.0.9 httplib2-0.22.0 httpx-0.28.1 idna-3.10 jsonpatch-1.33 jsonpointer-3.0.0 jsonschema-4.23.0 jsonschema-specifications-2025.4.1 langchain
-0.3.25 langchain-core-0.3.59 langchain-google-genai-2.0.10 langchain-text-splitters-0.3.8 langsmith-0.3.42 narwhals-1.36.0 numpy-2.2.5 orjson-3.10.1
8 packaging-24.2 pandas-2.2.3 pdfminer.six-20250327 pdfplumber-0.11.6 pillow-11.2.1 proto-plus-1.26.1 protobuf-5.29.4 pyarrow-20.0.0 pyasn1-0.6.1 pya
sn1-modules-0.4.2 pycparser-2.22 pydantic-2.11.4 pydantic-core-2.33.2 pydeck-0.9.1 pyparsing-3.2.3 pypdfium2-4.30.1 python-dateutil-2.9.0.post0 pytho
n-dotenv-1.1.0 pytz-2025.2 referencing-0.36.2 requests-2.32.3 requests-toolbelt-1.0.0 rpds-py-0.24.0 rsa-4.9.1 six-1.17.0 smmap-5.0.2 sniffio-1.3.1 s
treamlit-1.44.1 tenacity-9.1.2 toml-0.10.2 tornado-6.4.2 tqdm-4.67.1 typing-inspection-0.4.0 typing_extensions-4.13.2 tzdata-2025.2 uritemplate-4.1.1
 urllib3-2.4.0 watchdog-6.0.0 zstandard-0.23.0
(venv) ubuntu@ip-172-31-28-162:~/HemachandraHD_FinancialHealthAdvisor$ streamlit run streamlit_app.py --server.port 8501 --server.enableCORS false
2025-05-14 04:00:24.042
Warning: the config option 'server.enableCORS=false' is not compatible with 'server.enableXsrfProtection=true'.
As a result, 'server.enableCORS' is being overridden to 'true'.

More information:
In order to protect against CSRF attacks, we send a cookie with each request.
To do so, we must specify allowable origins, which places a restriction on
cross-origin resource sharing.

If cross origin resource sharing is required, please disable server.enableXsrfProtection.

Collecting usage statistics. To deactivate, set browser.gatherUsageStats to false.

  You can now view your Streamlit app in your browser.

  Local URL: http://localhost:8501
  Network URL: http://172.31.28.162:8501
  External URL: http://13.218.146.160:8501
master*
0 0